Winnower: Inferring Periodic Domains in Cellular Automata

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Abstract

Cellular automata (CAs) often generate periodic domains together with localized structures and domain interfaces. We present *Winnower*, a model-selection method for inferring a periodic spatiotemporal domain from a finite CA space–time observation. Each candidate temporal period and spatial displacement induces translation orbits over the observed sites. Orbit-majority states define the minimum-Hamming template, and Bernoulli normalized maximum likelihood (NML) ranks candidates by fit and complexity. Winnower returns the selected template and a defect mask. In preliminary one-, two-, and three-dimensional experiments, selected symmetries stabilized after short transients. In a sweep of 105 nontrivial Life-like rules, unpenalized likelihood selected the largest scanned period in every case, whereas NML selected period 1 for 97 rules. Several defect masks retained persistent localized structure. Winnower therefore provides a lightweight global preprocessing step for CA rule-space surveys before local particle or coherent-structure analysis. Data and code: `https://github.com/zitterbewegung/Winnower`.

Introduction

A cellular automaton (CA) updates a regular lattice synchronously under a local rule. Many rules produce space–time diagrams containing periodic domains, localized structures such as particles or gliders, and interfaces between domains. The domain supplies the background against which such structures are defined. We ask which temporal period and spatial displacement give the simplest adequate domain model for a finite observation.

Local causal-state methods reconstruct local spatiotemporal organization (Rupe and Crutchfield, 2024), whereas taxonomies, transient analyses, and compression-based approaches characterize rule-level dynamics (Vispoel et al., 2022; Rollier et al., 2024; Hudcová and Mikolov, 2021; Alfaro and Sanjuán, 2024; Cisneros, 2023). Winnower complements them by fitting global periodic domain templates with a complexity penalty suited to broad rule-space sweeps.

Method

Given a finite binary space–time block U, temporal period p, spatial displacement vector $\mathbf{s}$, and periodic lattice extents $\mathbf{D}$, Winnower fits templates satisfying $B[t{+}p,\ \mathbf{x}{+}\mathbf{s} \bmod \mathbf{D}] = B[t, \mathbf{x}]$. The translation partitions observed sites into orbits $\{O_j\}$. Assigning each orbit its majority state yields the minimum-Hamming template for that candidate; mismatches form the defect mask (fig. 1). Candidates are ranked by

$$\mathrm{NML}(p, \mathbf{s}) = \sum_j n_j\, H_b(\hat{\theta}_j) + \sum_j \mathcal{C}(n_j), \qquad (1)$$

where $n_j = |O_j|$, $\hat{\theta}_j = n_j^{(1)}/n_j$, H_b is binary entropy, and $\mathcal{C}(n)$ is the exact Bernoulli NML (Shtarkov) complexity for an n-observation class. The implementation evaluates $\mathcal{C}$ exactly for small classes and uses its leading asymptote $\frac{1}{2}\log_2(n\pi/2)$ for larger ones (Shtarkov, 1987; Grünwald, 2007; Yamanishi, 2023).

A compatible longer-period candidate can refine a shorter candidate's orbit partition without worsening Hamming disagreement or unpenalized likelihood, so fit-only selection favors longer periods. NML penalizes the additional orbit models. We formalized the corresponding divisibility and refinement result in Lean 4 (Achim et al., 2025). If candidates differ in per-site NLL rate, the linear fit gap eventually dominates the logarithmic penalty gap, stabilizing selection as the observation horizon grows.

Preliminary Results

We evaluated elementary CA Rules 30, 54, and 110; two-dimensional Life-like and totalistic rules; and three-dimensional Life-like rule B5/S45 (Bays, 1987, 2006). Figure 2 shows Rules 54 and 110. Rule 54 selected period 4 throughout, with its margin increasing from 681 to 2318 bits; B5/S45 selected period 1 throughout, increasing from 3324 to 7403 bits. Rule 110 changed from period 1 to its period-7 ether background, and S24/B11 changed from period 1 to period 2; both margins increased after the transition.

At $T{=}100$ in a sweep of 105 nontrivial Life-like rules, unpenalized NLL selected the largest scanned period for every

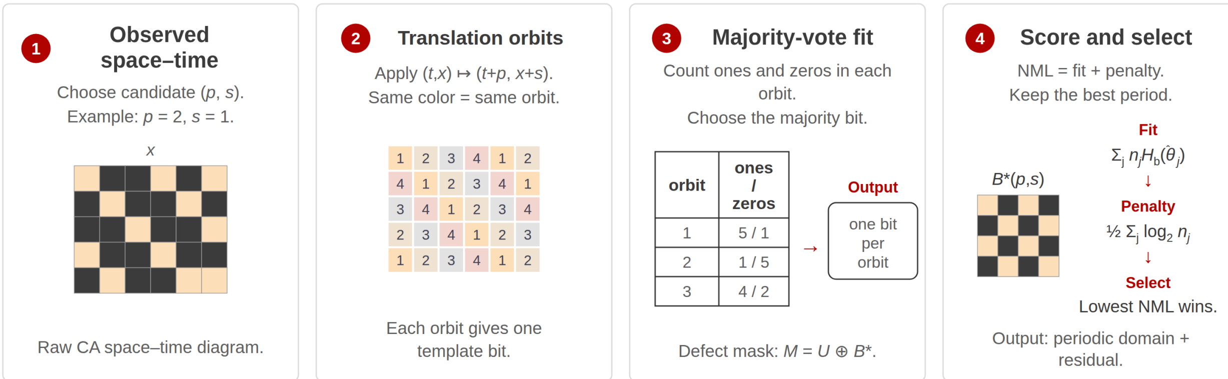

Figure 1: Winnower's pipeline. A candidate translation $(p, \mathbf{s})$ partitions observed space–time sites into orbits. Orbit majorities define the domain template, mismatches form the defect mask, and Bernoulli NML balances fit against model complexity.

rule, whereas NML selected period 1 for 97. Entropy-rate comparisons indicate that the inferred template captures a predictable component while the defect mask retains structured variation. In rules such as S37/B11, the defect population persisted and increased with lattice size (Seck-Tuoh-Mora et al., 2023). These residuals are candidate departures from the domain, not identified particles.

Reproducibility. The released commands and analysis scripts regenerate all reported results and fig. 2.

Discussion and Future Work

Winnower returns a global periodic domain template and defect mask. It assumes binary states, periodic boundaries, a supplied candidate set, and one global symmetry; it therefore complements rather than replaces local causal-state or particle analysis (Rupe and Crutchfield, 2024; Rupe et al., 2025). Future work will test sensitivity to observation length, boundaries, noise, and candidate-set size; connect residual masks to local tracking; and extend the method to multistate, continuous-state, and neural cellular automata (Hamon et al., 2024; Plantec et al., 2025; Niklasson et al., 2021).

Acknowledgements. We thank Editage for language editing. ChatGPT and Claude Code assisted with software development; Aristotle and Claude assisted with Lean proof development and verification.

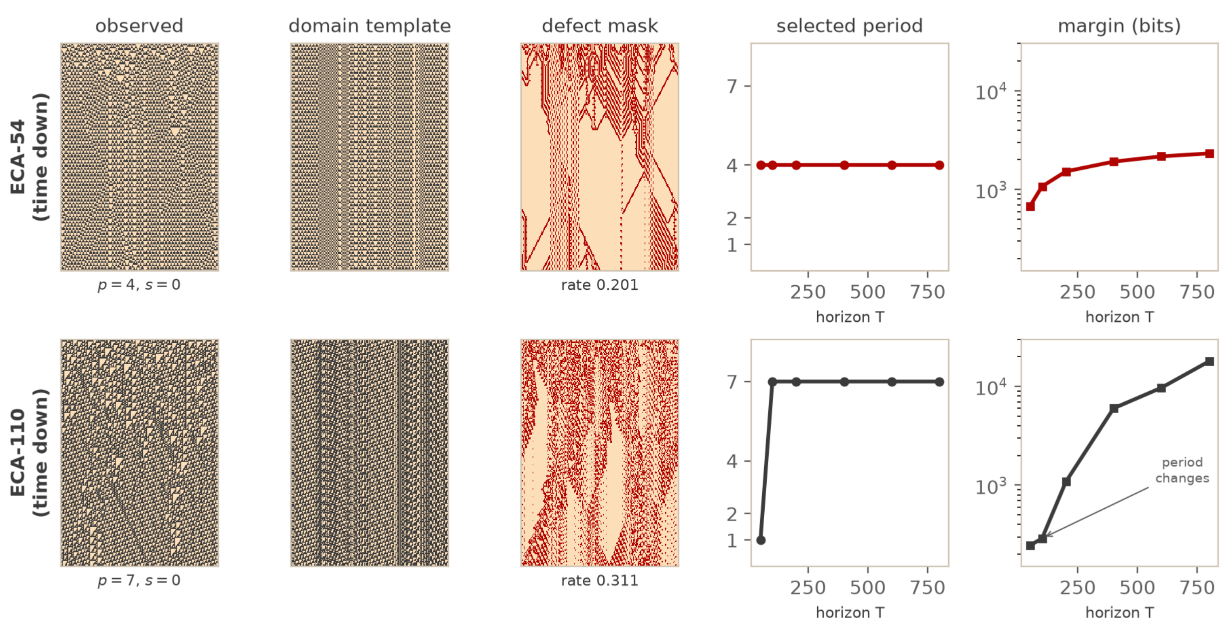

Figure 2: Elementary CA Rules 54 (top) and 110 (bottom). The left panels show the observation, selected shift-zero template, and defect mask; the right panels show selected period and NML margin versus horizon T. Rule 54 stays at period 4. Rule 110 changes once from period 1 to its period-7 ether domain, after which the margin grows.

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